

ORIGINAL INVESTIGATION



Effect of support foot placement on football instep kick performance

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ABSTRACT

Although the support foot plays an important role in kicking a football, there has been a paucity of research exploring the effect of the placement of the support foot on kicking performance. To investigate the kick performance under different support foot positions, ten male footballers were recruited to participate in two experiments: one determining the maximum ball speed and the second determining accuracy. The participants were instructed to plant their support foot on one of nine different spots marked in the form of a 3 × 3 shape on a piece of artificial grass and asked to kick the ball. In the first (maximum speed) test, the participants tried their best to kick the ball at the maximum ball speed from nine different support foot positions. In the second (accuracy) test, the participants kicked the ball toward the target area without restricting the support foot position. The ball speed, as well as the success rate, were recorded from each support foot position. Significantly higher ball speed and accuracy were obtained at medial positions than was the case at lateral positions from the nine spots. It was concluded that although footballers may choose different positions for support foot placement, the maximum ball speed and better accuracy could be expected when the support foot was next to or slightly in front of the ball centre without too much side-by-side separation (27–37 cm).

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Introduction

Kicking is the most studied and crucial skill in association football (also known as soccer or simply as football) to examine its various applications, such as passing, crossing, clearance, and shooting (Lees and Nolan 1998). For example, about 80% of goals were scored with kicking in the last three FIFA World Cups (Njororai 2013; Mülazimoğlu et al. 2015; Çobanoğlu 2019). Among the many types of kicking, the instep kick, which can produce a higher ball speed, is usually adopted for scoring purposes and is frequently analyzed in studies investigating the kicking of a football (Dørge et al. 2002; Nunome et al. 2002; Sterzing et al. 2009; Katis and Kellis 2010; Katis et al. 2015; Cerrah et al. 2020; Rabello et al. 2022). Kicking performance was typically determined based on ball speed and accuracy (Lees and Nolan 1998). While the former can be easily quantified, the latter lacks a standard evaluation procedure and is relatively understudied (Kellis and Katis 2007; Sterzing et al. 2009). In addition, there is a common agreement on the ‘speed-accuracy trade-off,’ suggesting that the ball speed will decrease significantly by about 20% when a higher demand is imposed on accuracy (e.g., kicking towards defined targets) (Asami 1983; Lees and Nolan 1998; Teixeira 1999; Sterzing et al. 2009).

The kicking process involves a complex sequence, including an angled approach, the support foot landing, sequential transfer of momentum in the kicking limb, foot-ball contact, and follow-through (Barfield 1998; Barfield et al. 2002). Due to playing the decisive role in generating power for the kicking leg in an open kinetic chain regardless of the kicking type (Lees et al. 2010), the support foot has also been argued to be a key

factor influencing the resultant ball speed and trajectory (Barfield 1998; Kellis et al. 2004; Harrison and Mannering 2006; Orloff et al. 2008; Inoue et al. 2014; Augustus et al. 2017). However, there has been a paucity of research exploring the effect of support foot placement on kicking performance (Lees et al. 2010). Previous research has investigated the support foot kinematics, kinetics, and ground reaction forces (Lees et al. 2010), or has generally considered the support foot position as a dependent variable when investigating the effect of approaching angle (Scurr and Hall 2009), gender (Orloff et al. 2008), dominant/non-dominant foot (McLean and Tumilty 1993; Harrison and Mannering 2006), and types of kicking (McLean and Tumilty 1993). Distance from a range of approximately 27 to 41 cm in the mediolateral and –5 to –11.7 cm in the anteroposterior directions has been reported in the literature. In addition, Kapidžić et al. (2014) reported a significant correlation between the distance (behind the ball) and the ball speed, suggesting that a smaller posterior distance results in a higher ball speed. This was attributed to the larger radius of gyration of the more extended kicking leg when the support leg is far from the ball, causing a smaller angular leg swing velocity. Thus, a closer support foot position generates greater ball speeds.

As an indispensable factor for generating a powerful kick, the support foot placement position has lacked thorough discussion in the literature. The purpose of the current study was to investigate the effect of support foot placement on kick performance. Since it was previously reported that the maximum ball velocity and accuracy were the main factors to quantify the kick performance, two experiments are conducted

in this regard. The first one examined the ball speed generated with different prescribed support foot placement locations; the second one focused on the success rate under accuracy demand conditions. While the authors initially planned to have the same experimental setting (i.e. prescribed support foot position) for both experiments, our pilot test revealed that almost none of the participants could shoot the ball under accuracy demand when planting their foot at some unfamiliar support foot position. Thus, only the participants' preferences on support foot positions and the success rate were observed in the second experiment. We hypothesized that an optimum support foot position would generate higher ball speed and accuracy during football instep kicking.

Methods

Ten male first division collegiate football players (age: 21.90 ± 1.20 years, height: 1.77 ± 0.06 m, weight: 66.20 ± 7.57 kg) without any critical lower extremity injuries reported in the previous six months were recruited in the current study. All participants had received more than two years of regular football training. The participants wore standard football cleats (Tiempo Genio II Leather-FG, Nike Inc., Beaverton, OR) to eliminate the effect of shoe preference on ball speed (Sterzing & Hennig, 2008) and accuracy (Hennig, Althoff, & Hömme, 2009), and FIFA approved soccer balls (F5V5000, Molten, Japan) were used throughout the study. Prior to conducting this research, approval from the National Cheng Kung University Governance Framework for Human Research Ethics was obtained, and an informed consent statement was signed by each participant.

The entire study comprised two experiments. The first experiment (the maximum ball speed test) was conducted in the laboratory. A piece of $2 \text{ m} \times 1 \text{ m}$ artificial grass carpet was placed on the floor to mimic the football pitch surface and provide adequate friction. It was attached to the floor using duct tape, with non-slip pads underneath to prevent sliding. In the current study, no infill was applied to the artificial grass. A total of nine different spots were marked within a 3×3 shaped area ($30 \text{ cm} \times 30 \text{ cm}$) on the artificial grass with white tape to indicate the positions at which the participants would plant their support foot (Figure 1). The centre of the nine support foot positions (spot 5) was marked 37 cm to the left of the centre of the ball, and the others were equally separated

by 10 cm in both the anteroposterior and mediolateral direction. These positions were not normalized according to body height or shoe size due to the insignificant effect reported previously (Cockcroft and Van Den Heever 2016). The selection of this setting followed both the previous literature and the results of our pilot study. With an approach of approximately two to three steps and an angle of 45° , the participants were instructed to plant the support foot roughly in the centre of each of the spots and asked to kick the ball at the highest ball speed possible (without any accuracy restriction) towards a net placed 5 m apart from the ball. A digital radar gun (R2000, JUGS Sports, Tualatin, OR) with an accuracy of ± 0.5 mph (Okholm Kryger et al. 2019) was set behind the net to monitor the ball speed. The participants shot five successful trials consecutively from each spot. The sequence of the support foot placement of each participant was randomized to prevent the effect of learning. After completing five trials from the same spot, they were substituted by the next participant to finish the same task to provide adequate rest time and prevent the effect of fatigue. To summarize, a total of 45 successful trials were collected in the ball speed test. The trials were only considered to be successful if both of the experimenters (who stood at both sides) confirmed the participants had planted the centre of their support foot (i.e., approximately at midfoot, the section of the foot composed of the cuboid, navicular, and cuneiform bones) on the correct spot.

The second experiment (accuracy test) was conducted on a football pitch. Two participants quit after completing the first experiment, and thence, only eight participants completed the second test. The participants were instructed to shoot 20 penalty kicks (12 yards apart) towards two target zones ($2.44 \text{ m} \times 1.50 \text{ m}$), which were indicated using two sticks placed at the left and right part of a standard football goal, in alternating order (Figure 2). The nine support foot positions were marked on the pitch surface with the same dimensions as in the former test. During the pilot study, it was found that the accuracy and the success rate were severely affected if the participants were strictly instructed to plant their feet on some specific spots. Therefore, the participants were only asked to shoot the ball to the target zone as fast as they could (to mimic the situation that they normally shoot a penalty in the competition) without any restriction on the support foot placement. Two experimenters (first-division collegiate footballers given adequate training and practice time for this study) stood at both sides to visually

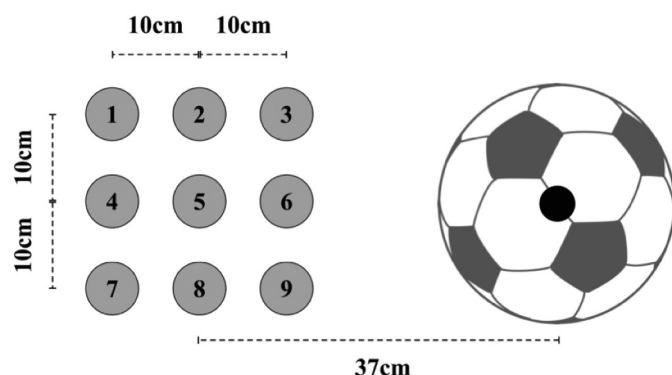


Figure 1. Illustration of support foot placement. (Not to scale).

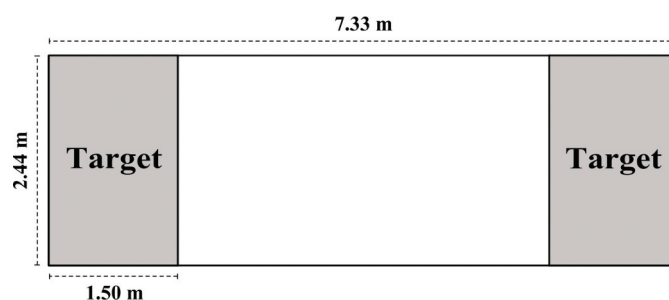


Figure 2. Illustration of the target zone in the accuracy experiment.

judge and record whether the ball successfully passed through the target zone, as well as the support foot position in every trial. If the two experimenters did not have an agreement on the spot where the participant planted their support foot, the participant was asked to repeat the trial. In addition, a digital radar gun was set behind the goal to monitor the ball speed. Participants were asked to retake the trial if the ball speed did not exceed 16.67 m/s (60 kph) (Savelsbergh et al. 2002) to prevent them from intentionally shooting the ball at an abnormally slow speed to achieve higher accuracy.

All statistical analyses were performed using SPSS 17.0 (International Business Machines Corporation, Armonk, NY). For the first experiment, a one-way repeated-measures ANOVA was used to examine the effect of different support positions on the resultant ball speed. Simple main effects were examined with Bonferroni adjusted p values to evaluate significant differences. For the second task, a chi-squared test was used to evaluate the preference for the support foot placement under accuracy demand. The significance level was set as $\alpha = .05$.

Results

The mean and standard deviation of ball speed obtained from the nine support foot positions in the first experiment are shown in Figure 3. Significant differences in ball speed among these foot positions were found ($F_{(8,72)} = 7.858$, $p < .001$, $\eta^2 = .47$). The post-hoc test revealed that significantly lower

ball speeds were found in spots 1, 4, 7, and 8. Furthermore, spots 1, 4, and 7 (the farthest lateral positions from the ball) resulted in a significantly slower speed than the other spots (i.e., spots 2, 3, 5, 6, and 9). Ball speed at spot 8 was only slower than spot 2, 3, and 5. For the second experiment, statistically different preferences were also found for the support foot position ($\chi^2_{(7)} = 112.4$, $p < .001$, $\phi_c = .30$). Among the 160 trials, the participants tended to plant their feet in spots 5, 6, 8, and 9 (right lower quadrant) (Table 1). Higher success rates (than the overall success rate) were obtained in spots 2, 3, 5, and 6 (right upper quadrant). Nevertheless, it should be noted that according to our experimental log, 12 out of 13 attempts from spot 2 were obtained from one particular participant, another participant made 7 out of 21 attempts from spot 9.

Discussion

Although the support leg placement is considered to be a key factor in the kinetic chain contributing to the generation of kicking movements, research in the past focused primarily on the kicking leg. Previous studies have generally examined maximum ball speed and/or accuracy with different independent variables such as the approach angle (Scurr and Hall 2009), type of kick (Alcock et al. 2012), dominant foot (McLean and Tumilty 1993; Harrison and Mannering 2006), etc. Thus, the support foot placement was only reported as an outcome. Based on the literature review and personal communications with football coaches and athletes, attempting to maximize ball speed may

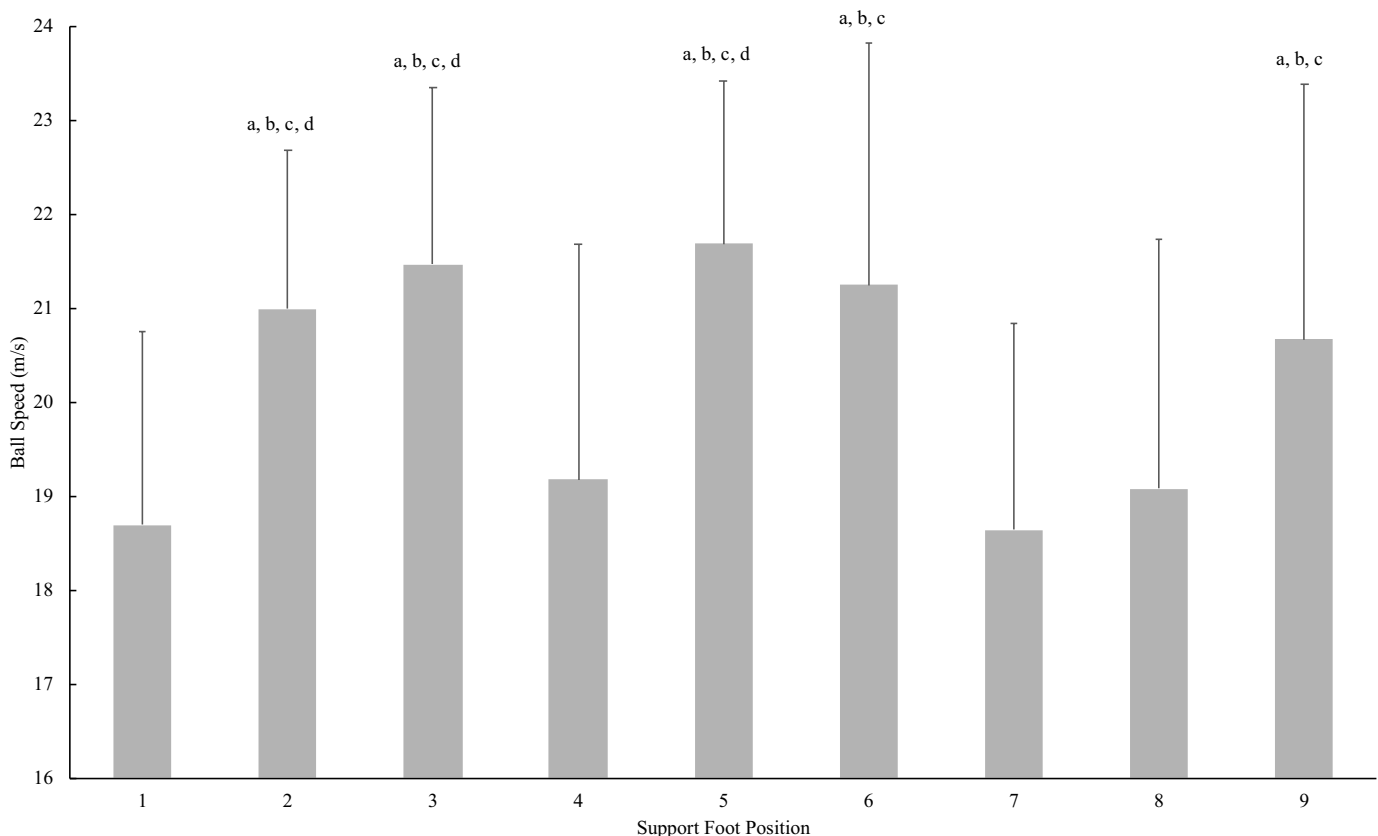


Figure 3. Ball speed from the nine support foot positions obtained from the first experiment. Note. The letters a, b, c, and d indicated significant differences in ball speed compared with placing the support foot position at the spots 1, 4, 7, and 8, respectively.

Table 1. Number of shots attempted and success rates for different support foot positions.

Position	Count (shots attempted)	Count (shots on target)	Success rate
1	1	0	0%
2	13	8	61.50%
3	6	3	50%
4	2	0	0%
5	51	31	60.78%
6	30	14	46.67%
7	0	0	0%
8	36	10	27.78%
9	21	8	38.10%
Total	160	74	46.25%

result in different approaching/kicking movements as opposed to those when attempting to achieve accuracy. Therefore, instead of including both the maximum ball speed and accuracy as the criteria to quantify the kicking performance, in the current study, two separate experiments were conducted to specifically investigate the effect of the support foot placement on ball speed and accuracy (rather than simply recording the observed support foot position).

In the first experiment, the support foot position was systematically controlled in each trial, and the resultant ball velocity was observed at each spot. Since the ball was kicked for a shorter distance without a demand for accuracy, the participants were able to kick the ball when they planted their support foot at every spot, though some spots resulted in significantly lower ball speed. The second experiment was aimed toward investigating the effects of different support foot placement locations under accuracy demand conditions. The experimental setting employed a common penalty scenario with specific targets marked on the goal, which is similar to Li et al. (2015) and Scurr and Hall (2009). During the pilot testing of the experimental setup, however, most participants reported difficulty in shooting the ball to the target area when planting their foot at unfamiliar support foot positions, especially at spots 1, 4, 7 (the farthest spots to the ball). Thus, we decided to employ the current setup to collect their preferences on support foot positions and the success rate. It is also supported by the current results (Table 1) that only three shots were attempted on the farthest spots (spot 1, 4, 7) with a 0% of success rate, which revealed the difficulty for the participants to perform a penalty kick accurately with that lateral distance. Thus, allowing the participants to perform kicking in their preferred way in the second experiment might provide a better insight into the relationship between different relative support foot placement and kicking performance.

An average maximum ball speed of 21.67 ± 1.73 m/s was recorded in the first experiment. Although a relatively slower ball velocity was obtained, which may have been due to the shorter approaching distance allowed by the artificial grass carpet, the velocity was still in a reasonable range based on previously reported velocities of 20.41 ± 2.44 m/s (Isokawa 1988), 20.14 ± 1.58 m/s (Kellis et al. 2004) and 22.70 ± 3.10 m/s (Orloff et al. 2008). The results of our first experiment revealed that different support foot placements did affect maximum ball speed. Overall, the participants produced higher ball speeds when placing their support foot at the right upper quadrant of

the nine chosen spots in the current study (spots 2, 3, 5, and 6). More specifically, the fastest ball speed occurred at spot 5, followed by spot 3 and spot 6 (i.e., 37 cm/0 cm, 37 cm/10 cm, and 27 cm/0 cm in mediolateral/anteroposterior direction apart from the centre of the ball, respectively). On the one hand, in the mediolateral direction, the ball speed with the farther foot placement relative to the ball (spots 1, 4, and 7) was significantly lower than that obtained from the centre (spots 2 and 5) and closer placement locations (spots 3, 6, and 9). On the other hand, relatively higher ball speeds were generated at the centre (spots 5 and 6) and more anterior placement locations (spots 2 and 3) compared to the posterior spots (8 and 9).

Several studies that investigated football instep kicking biomechanics also recorded the preferred support foot position for the participants (McLean and Tumilty 1993; Harrison and Mannering 2006; Orloff et al. 2008; Scurr and Hall 2009; Ball 2010; Li et al. 2015). Distance from a range of approximately 27 cm to 41 cm in the mediolateral and -5 cm to -11.7 cm in the anteroposterior directions was obtained. Compared to those studies, the foot placement spots where the higher ball speed occurred in the current study matched the previous observations in the mediolateral but not in the anteroposterior direction. This discrepancy may have firstly been due to the differences in the aim/setting in terms of the movement execution. That is, in most of the reviewed literature, a penalty scenario was simulated, where the most important goal of the participants was to score successfully (by kicking the ball into the goal or toward some specific targets) under both speed and accuracy demand conditions. However, the participants in the current study were told to kick the ball while attempting to generate the fastest ball speed possible while not considering any other demand. Therefore, they placed the support foot beside the ball centre in the anterior direction to generate a higher ball speed. This was also supported by Kapidžić et al.'s (2014) observation that a closer posterior support foot distance was significantly correlated to a higher ball speed. Secondly, in the penalty scenarios adopted previously, the participants were required to shoot penalty kicks with a distance of 10 m–11 m (12yds) between the ball and the target, but in the current experimental setting, the participants performed the kick towards a huge net from a distance of only 5 m. The shorter kicking distance to the target may have also affected the anteroposterior support foot distance. A similar experimental setting was adopted by Orloff et al. (2008). However, unfortunately, only the mediolateral distance was reported. In addition, inconsistent (or even unstated) definitions of the support foot position, including the distance between the ball and 1/3 of the foot length (measured from the toe tip), the 5th metatarsal, or the heel, have been employed (McLean and Tumilty 1993; Scurr and Hall 2009; Kapidžić et al. 2014; Li et al. 2015), which also increased the difficulty with comparing the results. Despite the inconsistent anteroposterior distances based on most of the previous results, Li et al. (2015) measured the distance from the support foot heel to the ball centre to be -5 cm $\pm$ 10 cm. Assuming the estimated foot length to be approximately 26 cm (de Leva 1996), their results were somewhat comparable to the current findings.

The results of the second experiment confirmed that football players have preferences for support foot placement when taking penalty kicks (under accuracy demand conditions). Approximately 86% of trials were taken with the support foot planted at the right lower quadrant of the nine spots (spots 5, 6, 8, and 9). Surprisingly, shots taken from the right upper quadrant (spots 2, 3, 5, 6) had higher success rates than those taken from spots 8 and 9. However, it should be noted that 12 out of 13 shots from spot 2, and 17 out of 21 shots from spot 9 were obtained by only two participants. Also, only 6 attempts were conducted from spot 3, although a success rate of 50% was obtained. After excluding spots that had been attempted by less than 4 participants, only spots 5, 6, and 8 met the criterion. Shots taken from spot 5 (37 cm/0 cm apart from the ball centre) obtained both the highest number of attempts and highest success rates, followed by spot 6 (27 cm/0 cm). Although 6 more attempts were counted at spot 8 (27 cm/-10 cm) compared to spot 6, the success rate decreased by almost half as compared to spot 6 (which was also below the average success rate).

Two studies with similar experimental settings obtained an average anteroposterior/mediolateral support foot distances of -9.7 cm to -11.7 cm/31.1 cm to 41 cm (depended on the approaching angle) (Scurr and Hall 2009) and an average of -5 cm/36 cm (Li et al. 2015), respectively. The preferred support foot placement for the penalty instep kick in the present study was similar to the results in both directions. However, taking the success rate into account, although the football players tended to plant their support foot slightly behind the ball centre in the anteroposterior direction, our results revealed that a smaller posterior distance (or even slightly in front of the ball centre) might increase accuracy since higher success rates at spots 5 and 6 were obtained than at spots 8 and 9. Nevertheless, the two outliers reported in the second experiment and the large standard deviations in the support foot posterior distance (from 5.3 cm to 10 cm) reported in the aforementioned studies also indicated that the support foot placement varied among the different football players, and the results suggest that this may be somewhat subject-specific.

Being the only body part that receives an external force (ground reaction force) before foot-ball contact during kicking (Kellis et al. 2004; Orloff et al. 2008; Katis and Kellis 2010), the support leg can significantly alter lower limb dynamics and affect kicking performance (Augustus et al. 2017). Based on the current results, the greatest ball velocity and accuracy appeared to occur at an anteroposterior/mediolateral distance of approximately 0–10/27–37 cm relative to the ball centre (Figure 4), which confirmed our hypothesis that an optimum support foot position existed that would lead to better instep kick performance. Another critical finding was that a lateral support foot position farther than 37 cm (i.e., spots 1, 4, and 7) resulted in significantly worse performance. It was proposed that kicking with a wider foot placement requires different joint dynamics and movement patterns (compared to kicking with the optimum foot placement) to compensate for the extra distance to the ball.

During maximal instep kicking, the entire proximal-to-distal body segment motion is aimed towards maximizing the kicking leg (shank/foot) velocity at the time of ball contact (Dørge et al.

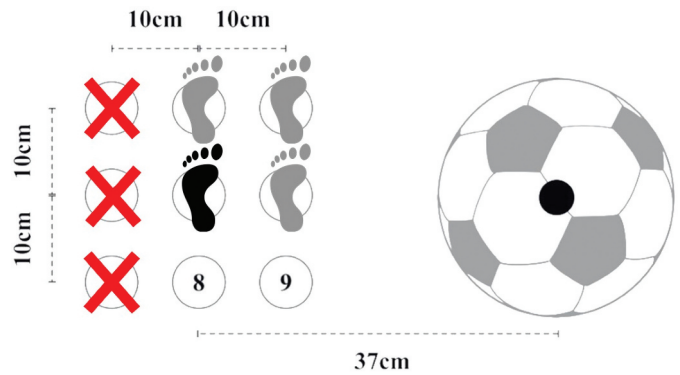


Figure 4. Illustration of the optimum support foot position for the instep kick. (Not to scale) Significantly better kicking performance was obtained with the support foot planted at a range of 0 cm–10 cm/27 cm–37 cm (spots 2, 3, 5, and 6), with the highest ball speed and accuracy occurring at 0 cm/37 cm (spot 5). The worst kicking performance was obtained with the support foot placed farther than 37 cm laterally to the ball (spots 1, 4, and 7).

2002). Previous studies highlighted that ‘lifting the body upward’ plays an important role in accelerating the downswing of the kicking leg (Inoue et al. 2000, 2014; Nunome and Ikegami 2005; Lees et al. 2009; Augustus et al. 2017). For instance, greater hip vertical displacement and velocity were observed for the expert football players and proved to be useful in increasing the swing speed by computer simulation (Inoue et al. 2000), and this vertical lift was proposed to be associated with the extension of the support leg knee and hip (Nunome and Ikegami 2005). The support leg may contribute to kick performance by lifting the body and adding to the vertical velocity of the foot at impact (Lees et al. 2009). The initial support leg knee flexion also has a role in absorbing the shock of landing, and the following active contraction power and extension moment of the support leg passively contributes to accelerating the kicking foot towards the ball (Inoue et al. 2014; Augustus et al. 2017). Therefore, it is reasonable to assume that a wider foot placement may have reduced the effectiveness of the extension of the support foot knee and hip, in turn negatively affecting the motion-dependent moment for accelerating the kicking leg, resulting in worse performance. In addition, an unstable body and unfamiliar movement pattern due to a wider foot placement may also decrease the quality of the foot contact to the ball (Andersen 1999; Nunome et al. 2006) or cause an offset distance between the impact point and the centre of the ball (which increase the ball spin) (Asai et al. 2002; Kellis and Katis 2007), resulting in decreased ball velocity and worse accuracy.

There are several limitations of the current study. First, in the second experiment, participants were instructed to shoot towards the left and right target in random order. The current study did not record the foot placements in each target direction separately. Since differences in foot placement when aiming towards different directions have been observed (Li et al. 2015), reporting the foot placements in separate kicking directions will be desired in future studies. Second, the correctness of the support foot placement was subjectively determined by two experimenters during data collection. Ball (2012) reported a support foot stance time of 0.27 s during maximal kicks. Although the time was considered to be sufficient for

identifying foot position (DiCarlo et al. 2012) and the experimenters were experienced footballers with refereeing experience and trained and practised before the experiment, visual judgment of the rapid movement may increase the chance of misjudging. Adopting a video-based method for reducing human error is suggested in future studies. The experimenters' judgment could also be improved by adding markers, such as coloured ink or tape to the outside of the shoes to identify the midfoot. Also, it is worth noting that although the chosen methodology was used to examine the effects of different support foot positions (e.g., farther away or closer to the ball centre), the investigation was accomplished by altering the foot positions among the prescribed spots instead of employing more accurate measurements (e.g., with a motion capture system). The current experimental design (without applying motion analysis) only allowed the outcome (i.e. ball velocity and accuracy) of the alteration of the relative support foot position to be observed, but was not able to determine the mechanism of how the altered foot placement affects the kick performance. The reported detrimental effect on kick performance at some spots could be caused by different factors, such as bad foot-ball contact, support/kicking leg and upper body dynamics, etc. (Barfield 1998; Kellis and Katis 2007; Lees et al. 2010). Nevertheless, the current study still provides a valuable reference for players who seek to improve their instep kicking performance, especially by altering their support foot in order to the optimum kicking performance. Further experiments are suggested to investigate the support leg dynamics when planting the foot at different positions relative to the ball.

Conclusions

The results of the current study provided new insights into how the support foot placement affects kicking performance by examining the instep kick executed from different positions. Overall, both of the experiments performed showed that the kicking performance of the participants was affected by planting the support foot at different positions (relative to the ball), in terms of altered resultant speeds and accuracy. A closer distance did not seem to significantly affect the performance of the participants, but a wider distance significantly decreased both the ball speed and accuracy. This is probably because wider foot placement may have reduced the effectiveness of the extension of the support foot knee and hip, in turn negatively affecting the motion-dependent moment for accelerating the kicking leg, resulting in lower distal segment speed and ball speed. While the preferred support foot distance between the ball and the support foot was consistent with that found in the previous literature, a discrepancy between the participants' habitual position and that producing the highest speed and success rate has been due to individual differences and the relatively small number of participants. Other conditions, such as a different target (goal) size and shot distance, may affect the planting of the support leg and the resultant performance, which suggests future research directions.

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References

- Alcock AM, Gilleard W, Hunter AB, Baker J, Brown N. 2012. Curve and instep kick kinematics in elite female footballers. *J Sports Sci.* 30(4):387–394.
- Andersen TB. 1999. Collisions in soccer kicking. *Sports Eng.* 2(2):121–125.
- Asai T, Carré MJ, Akatsuka T, Haake SJ. 2002. The curve kick of a football I: impact with the foot. *Sports Eng.* 5(4):183–192.
- Asami T. 1983. Analysis of powerful ball kicking. *Biomechanics.* VIII-B:695–700.
- Augustus S, Mundy P, Smith N. 2017. Support leg action can contribute to maximal instep soccer kick performance: an intervention study. *J Sports Sci.* 35(1):89–98.
- Ball K. 2010. Kick impact characteristics for different rugby league kicks. Kick impact characteristics for different rugby league kicks. In: *Proceedings of 28th International Conference on Biomechanics in Sports.* Michigan, USA.
- Ball K. 2012. Application of biomechanical kicking research. In: *Proceedings of 30th International Conference on Biomechanics in Sports Melbourne, Australia.*
- Barfield WR. 1998. The biomechanics of kicking in soccer. *Clin Sports Med.* 17(4):711–728.
- Barfield WR, Kirkendall DT, Yu B. 2002. Kinematic instep kicking differences between elite female and male soccer players. *J Sports Sci Med.* 1(3):72.
- Cerrah AO, Şimsek D, Soylu AR, Nunome H, Ertan H. 2020. Developmental differences of kinematic and muscular activation patterns in instep soccer kick. *Sports Biomech.* 1–16.
- Çobanoğlu HO. 2019. Analysis of goal scored on Russia world cup 2018. *J Educ Training Stud.* 7(2):184–191.
- Cockcroft J, Van Den Heever D. 2016. A descriptive study of step alignment and foot positioning relative to the tee by professional rugby union goal-kickers. *J Sports Sci.* 34(4):321–329.
- de Leva P. 1996. Adjustments to Zatsiorsky-Seluyanov's segment inertia parameters. *J Biomech.* 29(9):1223–1230.
- DiCarlo JJ, Zoccolan D, Rust NC. 2012. How does the brain solve visual object recognition? *Neuron.* 73(3):415–434.
- Dørge HC, Andersen TB, Sørensen H, Simonsen EB. 2002. Biomechanical differences in soccer kicking with the preferred and the non-preferred leg. *J Sports Sci.* 20(4):293–299.
- Harrison AJ, Mannering A. 2006. A biomechanical analysis of the instep kick in soccer with preferred and non-preferred foot. In: *Proceedings of the 24th International Symposium on Biomechanics in Sports Salzburg, Austria.*
- Hennig E M, Althoff K and Hoemme A. (2009). Soccer footwear and ball kicking accuracy. *Footwear Science,* 1(sup1), 85–87.
- Inoue SI, Ito T, Sueyoshi Y, O'Donoghue RK, Mochinaga M. 2000. The effect of lifting the rotational axis on swing speed of the instep kick in soccer. In: *Proceedings of the 18th International Symposium on Biomechanics in Sports Hong Kong, China (pp.).*
- Inoue K, Nunome H, Sterzing T, Shinkai H, Ikegami Y. 2014. Dynamics of the support leg in soccer instep kicking. *J Sports Sci.* 32(11):1023–1032.
- Isokawa M. 1988 A biomechanical analysis of the instep kick motion in soccer Reilly, T, Lees, A, Davids, K, Murphy, W J. . Science and football, (London: E & FN Spon) 449–455 .
- Kapdžić A, Huremović T, Biberović A. 2014. Kinematic analysis of the instep kick in youth soccer players. *J Hum Kinet.* 42(1):81–90.
- Katis A, Kellis E. 2010. Three-dimensional kinematics and ground reaction forces during the instep and outstep soccer kicks in pubertal players. *J Sports Sci.* 28(11):1233–1241.
- Katis A, Kellis E, Lees A. 2015. Age and gender differences in kinematics of powerful instep kicks in soccer. *Sports Biomech.* 14(3):287–299.

- Kellis E, Katis A. **2007**. Biomechanical characteristics and determinants of instep soccer kick. *J Sports Sci Med*. 6(2):154.
- Kellis E, Katis A, Gissis I. **2004**. Knee biomechanics of the support leg in soccer kicks from three angles of approach. *Med Sci Sports Exerc*. 36(6):1017–1028.
- Lees A, Asai T, Andersen TB, Nunome H, Sterzing T. **2010**. The biomechanics of kicking in soccer: a review. *J Sports Sci*. 28(8):805–817.
- Lees A, Nolan L. **1998**. The biomechanics of soccer: a review. *J Sports Sci*. 16(3):211–234.
- Lees A, Steward I, Rahnama N, Barton G. **2009**. Lower limb function in the maximal instep kick in soccer. In: *Contemporary sport, leisure and ergonomics*. London: Routledge; p. 149–159.
- Li Y, Alexander MJ, Glazebrook CM, Leiter J. **2015**. Prediction of kick direction from kinematics during the soccer penalty kick. *Int J Kinesiol Sports Sci*. 3(4):1–7.
- McLean B, Tumilty D. **1993**. Left-right asymmetry in two types of soccer kick. *Br J Sports Med*. 27(4):260–262.
- Mülazimoğlu O, Afyon YA, Hazar K, Yanar Ş, Dalli M, Isikdemir E. **2015**. The analysis of the goals scored in round of 16 in FIFA 2014 world cup. *J Educ Sociol*. 6(2):85, 87.
- Njororai W. **2013**. Analysis of goals scored in the 2010 world cup soccer tournament held in South Africa. *J Phys Educ Sport*. 13(1):6.
- Nunome H, Asai T, Ikegami Y, Sakurai S. **2002**. Three-dimensional kinetic analysis of side-foot and instep soccer kicks. *Med Sci Sports Exerc*. 34(12):2028–2036.
- Nunome H, Ikegami Y. **2005**. The effect of hip linear motion on lower leg angular velocity during soccer instep kicking. In: *Proceedings of the 23rd Symposium on Biomechanics in Sports Beijing, China*.
- Nunome H, Lake M, Georgakis A, Stergioulas LK. **2006**. Impact phase kinematics of instep kicking in soccer. *J Sports Sci*. 24(1):11–22.
- Okholm Kryger K, Mitchell S, Forrester S. **2019**. Assessment of the accuracy of different systems for measuring football velocity and spin rate in the field. *Proc Inst Mech Eng Part P: J Sports Eng Technol*. 233(2):324–330.
- Orloff H, Sumida B, Chow J, Habibi L, Fujino A, Kramer B. **2008**. Ground reaction forces and kinematics of plant leg position during instep kicking in male and female collegiate soccer players. *Sports Biomech*. 7(2):238–247.
- Rabello R, Bertozzi F, Galli M, Zago M, Sforza C. **2022**. Lower limbs muscle activation during instep kick in soccer: effects of dominance and ball condition. *Sci Med Football* 6 1 . 40–48.
- Savelsbergh GJ, Williams AM, Kamp JVD, Ward P. **2002**. Visual search, anticipation and expertise in soccer goalkeepers. *J Sports Sci*. 20(3):279–287.
- Scurr J, Hall B. **2009**. The effects of approach angle on penalty kicking accuracy and kick kinematics with recreational soccer players. *J Sports Sci Med*. 8(2):230.
- Sterzing T and Hennig E M. (2008). The Influence of Soccer Shoes on Kicking Velocity in Full-Instep Kicks. *Exercise and Sport Sciences Reviews*, 36(2), 91–97.
- Sterzing T, Lange JS, Wächtler T, Müller C, Milani TL. **2009**. Velocity and accuracy as performance criteria for three different soccer kicking techniques. In: *Proceedings of the 27th International Society of Biomechanics in Sports Conference Limerick, Ireland*.
- Teixeira LA. **1999**. Kinematics of kicking as a function of different sources of constraint on accuracy. *Percept Mot Skills*. 88(3):785–789.

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